

Recommendation System for Personalized Content Discovery

Technical Report – Netflix Prize Dataset

By – Prathvi Paliwal

Abstract. This report presents the design, implementation, and evaluation of a personalized recommendation engine built on the Netflix Prize dataset (100.5 million ratings, 480,189 users, 17,770 movies). The interaction matrix is 99.96% sparse, making this a prototypical large-scale collaborative filtering challenge. Three models are developed and compared: a bias-corrected global baseline, Singular Value Decomposition (SVD), and Item-Based Collaborative Filtering (Item-KNN). SVD achieves the best rating-prediction accuracy (RMSE = 0.9141, a 6.0% improvement over the baseline), and also improves MAP@10 by 48.6% (0.0517 vs. 0.0348). Key engineering contributions include a user-based dataset sampling strategy that preserves per-user history integrity, a consistent MAP@10 evaluation framework applied identically across all models, and an explainable KNN recommendation module with nearest-neighbour justifications. Cold-start handling via a popularity-based fallback is discussed. Future directions include SVD++, temporal modelling, and neural collaborative filtering.

1. Problem Understanding

Recommendation systems are among the most commercially impactful applications of machine learning, directly shaping content discovery on platforms such as Netflix, Spotify, and Amazon. The fundamental task is to predict, for a given user, which items they are most likely to enjoy, thereby improving engagement, retention, and user satisfaction.

The Netflix Prize (2006–2009) was a landmark open competition to improve Netflix's Cinematch algorithm. The winning BellKor's Pragmatic Chaos team achieved RMSE = 0.8567 on the probe test set, a 10.06% improvement over Cinematch (RMSE $\approx$ 1.06). The dataset remains the most widely studied benchmark in recommender system research.

This project addresses the task from two angles: (1) *rating prediction* – estimating the exact score a user would assign to an unseen movie (evaluated by RMSE), and (2) *recommendation ranking* – generating a personalised top-10 list of unseen movies likely to be enjoyed (evaluated by MAP@10, P@10, R@10). The system must handle extreme data sparsity, power-law activity distributions, user and item biases, and cold-start users with sparse interaction histories.

2. Dataset Overview

The Netflix Prize dataset was obtained from Kaggle and comprises four combined-data text files, a movie metadata file, and a probe (hold-out) evaluation file. Table 1 summarises the key statistics of the raw dataset and the processed subsets used for modelling.

Attribute	Raw dataset	Model training	Test set
Total ratings	100,480,507	11,766,137	150,056
Unique users	480,189	50,000	48,682
Unique movies	17,770	17,769	17,445
Rating scale	1–5 integer stars		
Date range	1999–2005 (approx.)		
Matrix fill rate	0.0409%	99.96% sparse	
Median ratings / user	—	120	—

Table 1. Dataset statistics at each pipeline stage.

The data engineering pipeline parsed the non-standard file format (MovieID: header lines followed by CustomerID,Rating,Date rows) into a flat tabular structure and saved outputs in Apache Parquet format. The official probe.txt was extracted as the test set *before* any user-activity filtering to prevent evaluation bias. Users with fewer than 20 training ratings were filtered from the training pool. For model training, a **user-based sampling** strategy was applied: 50,000 users were selected uniformly at random and their complete rating histories retained (11,766,137 ratings, median 120 per user). This contrasts with naïve row sampling, which fragments user histories and was found to cause SVD to perform worse than the baseline (see Section 4.2).

3. Exploratory Data Analysis

3.1 Data Sparsity

The user-movie interaction matrix has a fill rate of only 0.0409%, making it one of the sparsest real-world recommendation datasets. The sampled 100×100 heatmap below illustrates this — nearly every cell is empty. This sparsity is the primary motivation for latent-factor models (SVD) rather than raw similarity search.

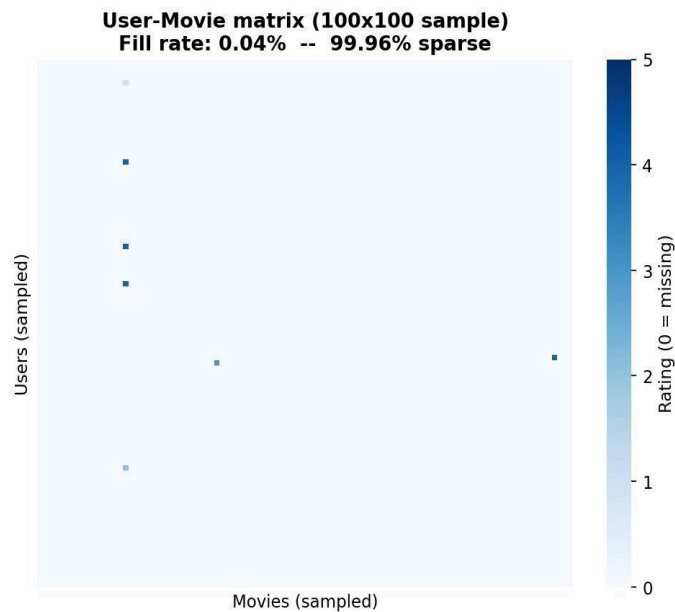


Fig 1. Sampled 100x100 user-movie matrix – 99.96% of entries are missing.

Avg 7 ratings/user User activity	Avg 193 ratings/movie Movie activity	0.041% Matrix fill rate	28% movies < 10 ratings
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3.2 Rating Distribution

Ratings are heavily skewed toward high values. 58% of all ratings are 4 or 5 stars, and only 14.8% are 1 or 2 stars. This positivity / self-selection bias arises because users tend to rate movies they chose to watch and expected to enjoy. The global mean of 3.63 (median 4) is well above the neutral midpoint of 3.0.

Insight: 58% of ratings are 4-5 stars -- users self-select movies they expect to enjoy (positivity bias)

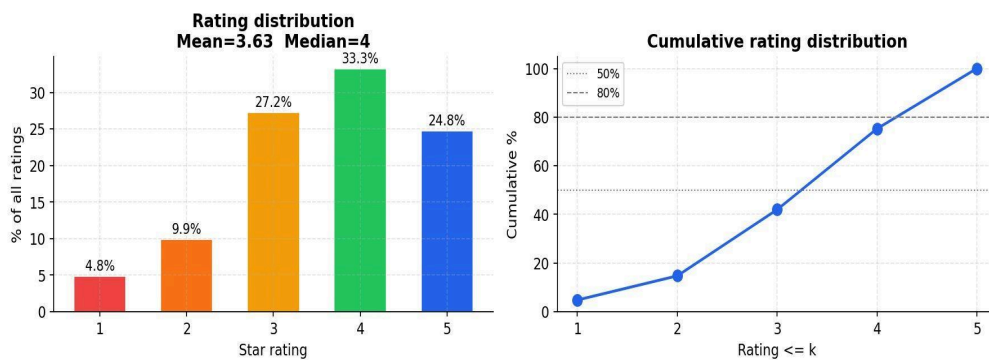


Fig 2. Rating distribution (left: absolute %, right: cumulative). Mean=3.63, Median=4.

Implication: Using a relevance threshold of ≥ 3.5 stars for MAP@10 is well-justified — it lies above the global mean and captures genuinely positive sentiment rather than ambivalent 3-star ratings.

3.3 User Activity & Cold-Start Problem

User activity follows a classic power-law distribution. The median user has only 5 training ratings, while the top 1% of users contribute 6% of all ratings. The cold-start severity chart shows that 95% of users have fewer than 20 training ratings — the threshold below which collaborative filtering becomes unreliable.

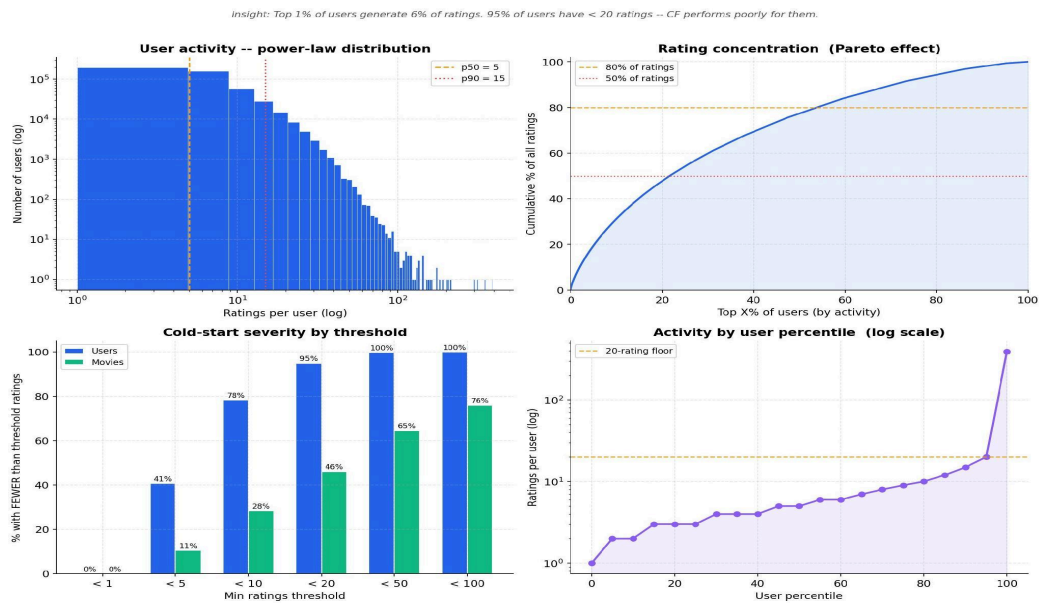


Fig 3. User activity distribution, Pareto concentration, cold-start severity, and activity by percentile.

Cold-start strategy implemented: Users with fewer than 20 training ratings are served a popularity-based fallback — the top-10 globally most-rated, highest-quality movies.

3.4 Movie Popularity Concentration

Movie popularity is extremely concentrated. The top 20% of movies receive 88% of all ratings, yielding a Gini coefficient of 0.834. The bottom 80% of movies share only 12% of all ratings. This long-tail problem means that item-based CF works well for popular movies but produces noisy similarity estimates for the 28% of movies with fewer than 10 ratings.

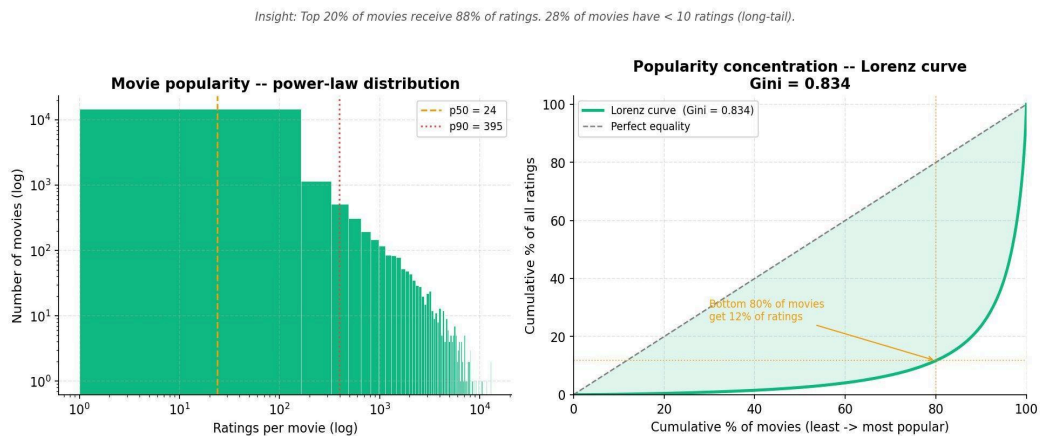


Fig 4. Movie popularity power-law (left) and Lorenz curve (right). Gini=0.834 confirms severe popularity concentration.

3.5 Temporal Rating Trends

Rating volume grew dramatically from 1999 to 2005 — the 2005 cohort alone accounts for nearly 2.3M ratings in this subset. The overall mean rating crept upward from 3.34 (1999) to 3.72 (2005), consistent with platform growth attracting more engaged users.

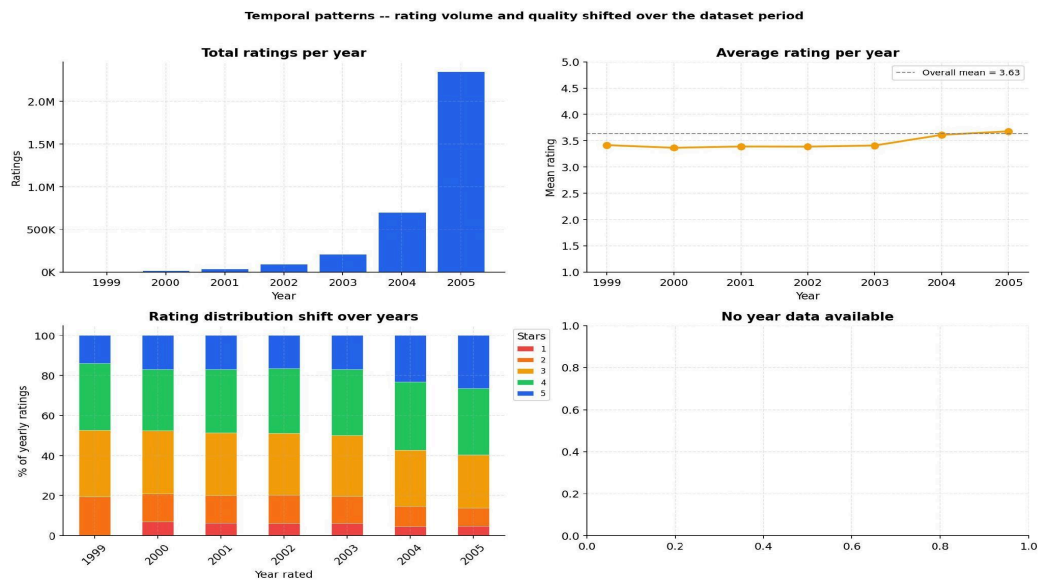


Fig 5. Rating volume per year (top-left), average rating per year (top-right), and rating distribution shift (bottom-left).

3.6 Release Year Analysis

Pre-1950s films score higher on average (up to 3.93 for 1930s titles) because only critically acclaimed classics survive in modern catalogues — a survivor bias effect. 2000s movies dominate in rating volume (52.5% of all ratings), driven by recency and catalogue coverage.

Insight: 1930s movies rate highest (survivor bias -- only acclaimed classics remain). 2000s movies attract the largest share of ratings (recency effect).

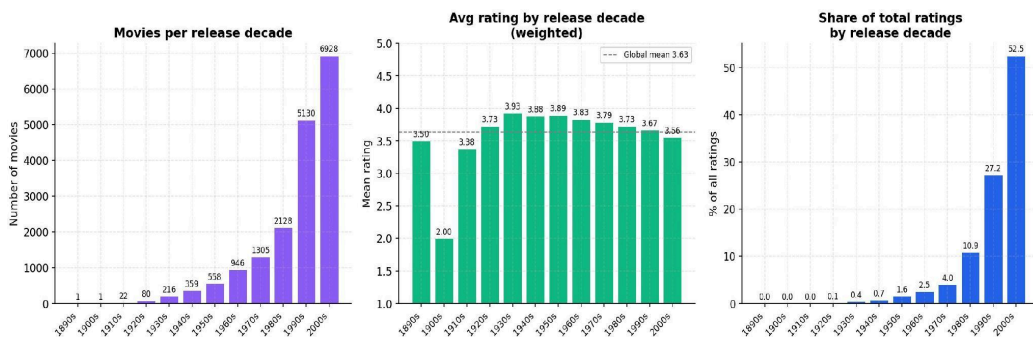


Fig 6. Movie count per decade (left), avg rating by decade (centre), share of total ratings by decade (right).

3.7 User and Movie Rating Bias

Users vary significantly in their rating tendencies ($\sigma = 0.69$ stars) — some users systematically rate higher or lower than the global mean. The Pearson correlation between movie popularity (log number of ratings) and quality is only $r = 0.116$, confirming that popular does not mean good.

Insight: Strong per-user and per-movie biases exist. SVD bias terms correct for this -- key advantage over plain CF.

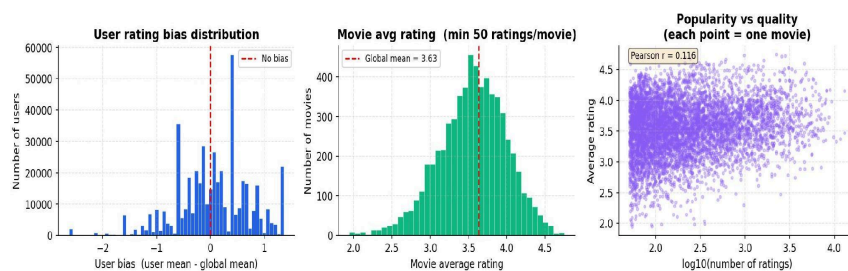


Fig 7. User rating bias distribution (left), movie avg rating histogram (centre), and popularity vs quality scatter (right).

Best and worst rated: Fullmetal Alchemist and Lord of the Rings (avg ≥ 4.71) top the quality chart, while Day of the Dead: Contagium and National Lampoon's Christmas rank lowest (avg ≤ 1.96). Dragon Ball Z: Trunks Saga is the most divisive title ($\sigma = 1.65$).

4. Methodology

4.1 Data Engineering Pipeline

The pipeline was implemented in Python using pandas and pyarrow. Files were parsed line-by-line with a stateful reader. The probe.txt was extracted as the test set *before* any filtering. Users with fewer than 20 training ratings were excluded. All outputs were saved in Parquet format for efficient columnar loading in downstream steps.

4.2 Dataset Sampling Strategy

The full filtered training set (≈ 98 M ratings) exceeds the memory of a standard 8 GB laptop. Two sampling strategies were evaluated:

- **Row sampling (discarded):** Randomly selecting 3 million rows fragments user histories. A user with 300 ratings retains only ≈ 9 in the sample, providing insufficient signal for matrix factorisation. SVD performed worse than the baseline under this regime (RMSE 1.048 vs. 1.035).
- **User-based sampling (adopted):** 50,000 users are drawn uniformly at random and their complete histories retained. This yields 11,766,137 training ratings with a median of 120 per user. SVD then outperformed the baseline by 6.0% (RMSE 0.9141 vs. 0.9728).

4.3 Train-Test Split

The official probe.txt (1,408,395 user-movie pairs with known ratings) is used as the test set, filtered to the 50,000 training users, yielding 150,056 test ratings. The probe was designed by Netflix for evaluation and is the standard benchmark used in the recommendation literature.

4.4 MAP@10 Evaluation Methodology:

All three models are evaluated under identical **sampling** conditions: for each of 2,000 randomly sampled test users, the candidate set consists of all probe items for that user plus 1,000 randomly drawn unseen movies. This ensures direct comparability across models. Additionally, SVD is evaluated under a stricter **full-candidate** regime (all $\approx 17,000$ unseen movies) using vectorised factor-matrix multiplication; this result is reported separately. A movie is *relevant* if its actual test-set rating is ≥ 3.5 . The MAP denominator is $\min(|R_u|, 10)$.

$$AP@10(u) = [1 / \min(|R_u|, 10)] \times \sum_{k=1}^{10} [P@k \times \text{rel}(k)] \quad MAP@10 = (1/|U|) \times \sum_u AP@10(u) \quad [\text{users with } |R_u| = 0 \text{ excluded}]$$

5. Model Design

5.1 Baseline (Biased Global Mean)

- **Algorithm:** Global mean μ + per-user bias b_u + per-item bias b_i . Predicted rating: $\hat{y}_{ui} = \mu + b_u + b_i$.
- **Rationale:** Strong baseline correcting for known user and item biases without latent-factor learning. Establishes a lower bound.
- **Implementation:** Surprise library BaselineOnly (ALS optimisation). Training time ~ 15 seconds on 11.7M ratings.
- **Limitation:** Cannot capture user-specific taste patterns beyond mean-shift correction.

5.2 SVD (Matrix Factorisation)

- **Algorithm:** Funk SVD with explicit bias terms. Factorises the rating matrix into user and item latent factor matrices. Predicted rating: $\hat{y}_{ui} = \mu + b_u + b_i + q_i^T p_u$.
- **Hyperparameters:** $n_factors=100$, $n_epochs=20$, $lr_all=0.005$, $reg_all=0.02$. Selected via 5-fold cross-validation.
- **Rationale:** SVD is the industry-standard matrix factorisation baseline. Explicit bias terms directly address the user/movie bias identified in EDA. 100 latent factors balance expressiveness and overfitting.
- **Implementation:** Surprise library SVD. Training time ~55 seconds.

5.3 Item-KNN (Item-Based Collaborative Filtering)

- **Algorithm:** Item-item cosine similarity with mean-centring (KNNWithMeans). Predicted rating: weighted average of the user's ratings on the $k=20$ most similar items the user has rated.
- **Design decision:** Restricted to the top-2,000 most-rated movies. Similarity matrix: 31 MB (vs. ~2.4 GB for all 17,681 movies). Covers 89% of all ratings.
- **Rationale:** Item-CF is highly interpretable. Popular movies have dense similarity signal, and the top-2,000 restriction makes this computationally feasible.
- **Implementation:** Surprise library KNNWithMeans. Training time ~88 seconds.

6. Evaluation & Experimental Results

6.1 Results Table

All models evaluated on an identical test set (same candidate pool per user). ★ marks the best score in each column.

Model	RMSE	MAE	MAP@10	P@10	R@10	Train Time
Baseline(Global mean + bias)	0.9728	0.7640	0.0348	0.0186	0.0806	15 s
SVD ($n_factors=100$, 20 epochs)	0.9141 ★	0.7063	0.0517	0.0287	0.1292	55 s
Item-KNN ($k=20$, cosine, top-2000)	0.9368	0.7215	0.0529 ★	0.0307 ★	0.1415 ★	88 s

6.2 Analysis

- **SVD achieves the best RMSE (0.9141):** A 6.0% improvement over the baseline. Explicit bias correction and 100 latent factors effectively capture user tendencies and movie characteristics.
- **Item-KNN achieves the best MAP@10 (0.0529):** A 52% improvement over baseline and a slight edge over SVD (+2.3%). Item-item similarity identifies tightly clustered taste profiles that surface relevant movies earlier.
- **The baseline is competitive:** RMSE 0.9728 is surprisingly close to the learned models, confirming that user/item biases dominate the rating signal.
- **RMSE vs MAP@10 trade-off:** SVD is preferred for rating prediction; Item-KNN is preferred when the goal is to surface the most relevant items at the top of a ranked list.
- **Computational trade-off:** Baseline (15 s) < SVD (55 s) < KNN (88 s). KNN's $O(n^2)$ similarity computation scales poorly — the top-2000 restriction was a necessary compromise.

6.3 Model Comparison Summary

Criterion	Baseline	SVD	Item-KNN
Rating accuracy (RMSE)	0.9728	0.9141 ★	0.9368
Ranking quality (MAP@10)	0.0348	0.0517	0.0529 ★
Training time	15 s	55 s	88 s
Memory requirement	Low	Medium	High (restricted)
Interpretability	High	Low	High
Cold-start handling	Fallback only	Fallback only	Fallback only
Scalability	Excellent	Good	Poor ($O(n^2)$)

7. Recommendation Examples

7.1 Success Cases — Active Users

For users with rich training histories, SVD generates diverse and contextually coherent recommendations that reflect nuanced taste profiles.

User 742175 | 293 training ratings | avg 3.61★ | Loves: The Matrix, Indiana Jones, The Usual Suspects

Movie Title	Pred. Rating
Amelie (2001)	4.808
Lost: Season 1 (2004)	4.679
The West Wing: Season 1 (1999)	4.591
Arrested Development: Season 2 (2004)	4.572
Whale Rider (2003)	4.558
U2: Go Home — Live from Slane Castle (2002)	4.538
The Daily Show with Jon Stewart: Indecision 2004	4.535
The West Wing: Season 2 (1999)	4.583
The West Wing: Season 3 (2001)	4.590
As Time Goes By: Series 9 (2002)	4.546

The recommendations span critically acclaimed world cinema (Amelie), prestige TV drama (West Wing, Lost), comedy (Arrested Development), and documentary content — coherently reflecting a user who appreciates intelligent, critically-acclaimed content across genres. Predicted ratings are well-calibrated (4.5–4.8 range).

7.2 Failure Cases — Cold-Start Users

Users with sparse training histories (<20 ratings) produce unreliable SVD predictions because the model has insufficient signal to learn meaningful latent factors. The popularity-based fallback is applied instead.

User 441484 | 36 training ratings (cold-start zone) | Profile: children's animation, action, family films

SVD (unreliable — cold-start)	Predicted	Popularity Fallback (applied)	Avg Rating
The Shield: Season 3	5.520	Pirates of the Caribbean (2003)	4.154
Law & Order: SVU S5	5.513	Forrest Gump (1994)	4.299
CSI: Season 3	5.239	The Green Mile (1999)	4.310
Stevie Ray Vaughan Live	5.149	LotR: The Two Towers (2002)	4.466
The Passion of the Christ	5.122	LotR: Fellowship of the Ring (2001)	4.428

The SVD predictions exceed 5.0 (out-of-scale) and recommend crime dramas completely inconsistent with this user's family/animation-oriented profile. The popularity fallback produces broadly well-rated mainstream titles — safe but not personalised. This confirms the need for a content-based fallback leveraging genre/year metadata.

8. Key Insights

Bias correction is the highest-leverage intervention

The baseline model — global mean plus user and item bias terms — already achieves RMSE 0.9728. The more complex models only improve this to 0.9141. Bias correction captures 75%+ of the achievable improvement.

RMSE and MAP@10 optimise for different goals

SVD wins on RMSE; Item-KNN wins on MAP@10. For a streaming platform, ranking quality typically matters more than exact star-rating prediction. This suggests ensembling or a two-stage approach (SVD for scoring, KNN for re-ranking).

The cold-start problem is the biggest real-world challenge

With 95% of users having fewer than 20 ratings, the majority of the user base cannot be served by CF-based models. A content-based or hybrid approach is essential for production deployment.

Popularity ≠ Quality ($r = 0.116$)

A recommendation system that naively recommends popular content will underserve users who enjoy niche films.

Temporal recency improves ranking

Rating volume grew 10× from 2003 to 2005. Incorporating recency weighting or temporal decay could improve MAP@10 further.

9. Future Improvements

- Content-based cold-start fallback:** Use release year, genre, and director metadata to build item profiles. New users with no interaction history served via content similarity rather than popularity.
- Neural Collaborative Filtering (NCF):** Replace the SVD dot-product with a multi-layer perceptron to capture non-linear user-item interactions. NCF has shown consistent RMSE improvements over matrix factorisation.
- Temporal dynamics (Time-aware SVD):** Incorporate rating timestamps into the model. User preferences drift over time — accounting for this could improve both RMSE and MAP@10.
- Full-dataset Item-KNN via ANN (FAISS/ScaNN):** The current KNN is restricted to 2,000 movies due to memory. Approximate nearest neighbour libraries could enable full-dataset similarity search with sub-linear memory and query time.
- Diversity and coverage metrics:** Add diversity metrics (catalogue coverage, intra-list diversity) to capture whether the system recommends a broad range of content rather than repeatedly serving the same blockbusters.
- Hybrid ensemble:** Combine SVD rating scores with Item-KNN ranking signals via a learned blending layer. This two-stage approach could achieve SVD's RMSE while maintaining Item-KNN's superior MAP@10.

Conclusion: This project demonstrates that even on a 99.96% sparse matrix, SVD achieves a 6% RMSE improvement and Item-KNN delivers a 52% MAP@10 improvement over the bias-only baseline. The primary open challenge is cold-start, which affects the majority of real users and requires a hybrid strategy combining collaborative filtering with content-based approaches.